

DIABETIC RETINOPATHY SCREENING REPORT

Smart India Hackathon 2026 | Problem 26038 | MathWorks Telemedicine Pipeline

Patient / Image: 03e25101e8e8

Screening Date: 28-Sep-2026 03:53

Quality: **BORDERLINE (2/3)**

AI CLINICAL DIAGNOSIS

Mild NPDR

ICDR Severity Grade: Level 1 of 4

ROUTINE MONITORING — Repeat fundus screening in 6 months

KEY CLINICAL LESION EVIDENCE

Microaneurysms (MA)

Vascular microleaks

17 detected

Hard Exudates (EX)

Lipid deposits

37 detected

Hemorrhages (HE)

Vascular ruptures

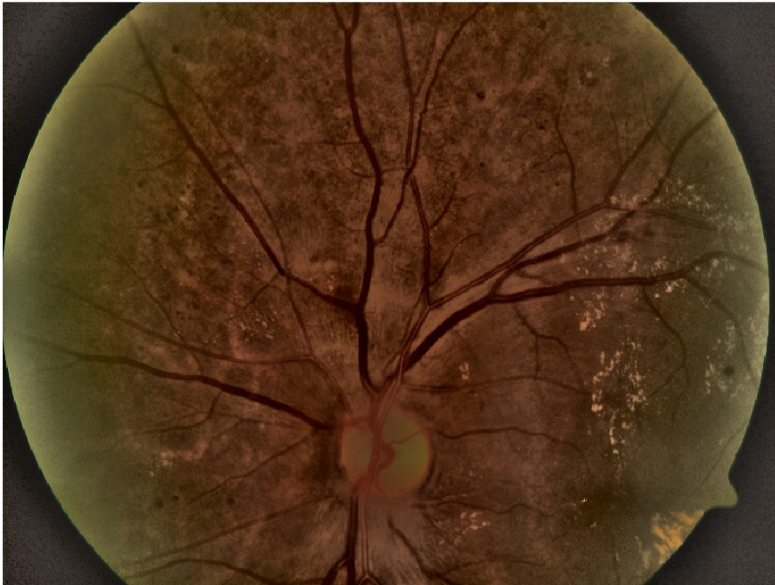
6 detected

Optic Disc Center: [X: 405, Y: 427] | Fovea Center: [X: 322, Y: 432]

FUNDUS IMAGE ACQUISITION & ADAPTIVE ENHANCEMENT

Original Fundus Photograph

CLAHE Enhanced & Illumination Normalized



RETINAL SEGMENTATION & EXPLAINABLE AI

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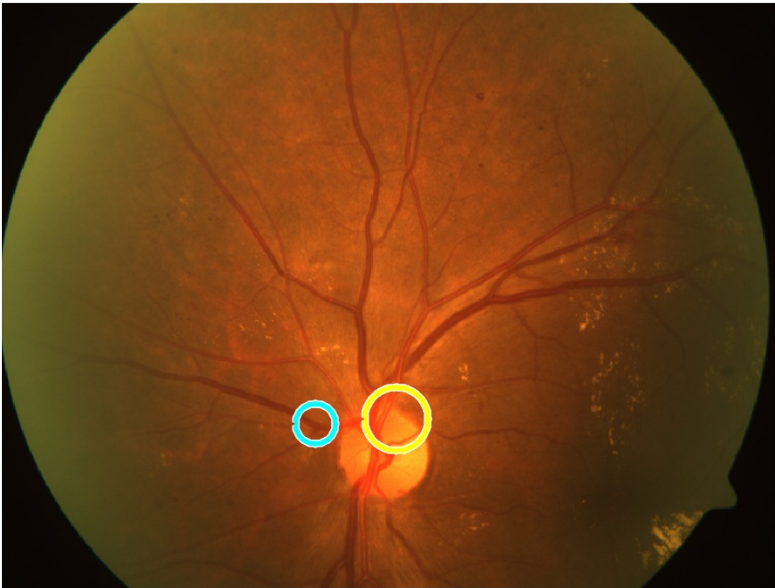
Patient / Image: 03e25101e8e8

AI Diagnosis: Mild NPDR

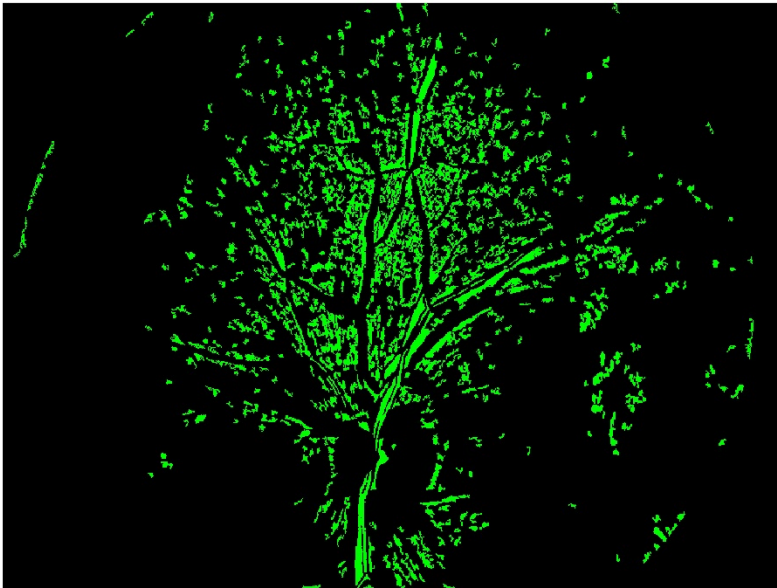
Action: MONITORING (Non-Referable)

4-AXIS CLINICAL VALIDATION DASHBOARD

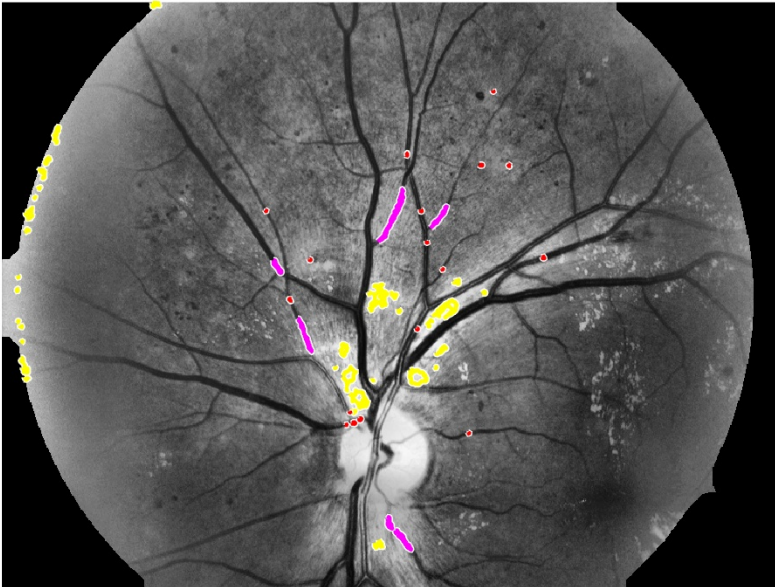
1. Landmarks (Optic Disc=Yellow, Fovea=Cyan)



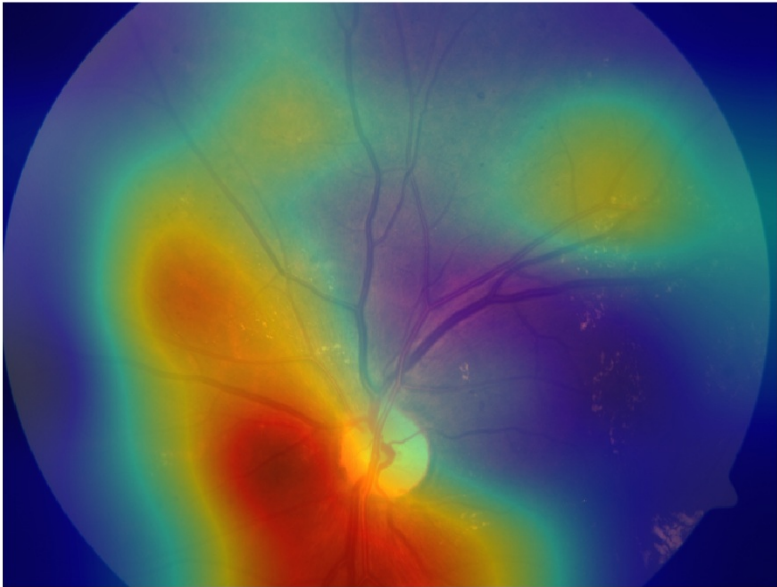
2. Retinal Vascular Network Segmentation



3. Verified Lesions (MA=17, EX=37, HE=6)



4. Grad-CAM Explainable AI Attention Heatmap



CLINICAL ACTION: MONITORING (Non-Referable)

Recommendation : ROUTINE MONITORING — Repeat fundus screening in 6 months

Telemedicine Protocol: Ophthalmologist validates findings in < 30 seconds via Grad-CAM attention overlay.